

Diagnosing Transformation-Order and Inverse-Domain Errors

Answer Key

Precalculus

Quick Answers

- | | |
|--|--|
| 1. (a) $g(2) = 9$, (a) $h(2) = 49$, — | 5. (a) value = 3, — |
| 2. (a) correct value = 1, — | 6. (a) x for $f(2x) = 8$, (a) x for $f(x) = 16$, — |
| 3. (a) $f(-4) = 16$, (a) the square root of $f(-4) =$ | 7. (a) correct value = 5, — |
| 4, — | 8. (a) correct value = -1 , — |
| 4. (a) correct value = 2, — | |
-

What is verified

11 of 19 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSF-BF.B.3 – *problems 1, 2, 4, 6, 7*

HSF-BF.B.4 – *problems 3, 5, 8*

Difficulty 1–5 of 5 across 19 tagged checks.

Common wrong answers

2. *If they got 2: shifted up by 1 before stretching, so the stretch doubled the 1 as well*
4. *If they got -6 : shifted down 4 before stretching, so the stretch tripled the 4*
7. *If they got -5 : reflected after adding 5, so the whole shifted graph was negated*
8. *If they got 5: kept the plus square root, which inverts the other branch of the parabola*
-

- 1.** $f(x) = x^2$; a student claims $g(x) = f(x) + 5$ and $h(x) = f(x + 5)$ have the same graph. (a) $g(2)$ and $h(2)$. (b) Name the error.

Solution (a): $g(2) = 2^2 + 5 = 9$

$g(2) = 9$

$$h(2) = (2 + 5)^2 = 49$$

$h(2) = 49$

Solution (b): *Instructor-judged.* The two values differ, so the claim fails — but the diagnosis is what is graded. The $+5$ in g sits *outside* f : it adds 5 to the output, so the graph moves up 5. The $+5$ in h sits *inside* f : it changes the input, so the graph moves 5 units *left*. Naming which is vertical and which is horizontal is the full answer.

2. $y = 2(x - 3)^2 + 1$ built in the wrong order. (a) True vertex height. (b) Name the error.

Solution (a): At $x = 3$ the squared term vanishes: $2(3 - 3)^2 + 1 = 0 + 1$.

$$y = 1$$

Solution (b): *Instructor-judged.* A vertical stretch multiplies whatever is already there, so it must be applied before the vertical shift. Shifting up 1 first and then stretching by 2 doubles that 1 and puts the vertex at height 2 — which is exactly the value the student reported. The horizontal shift of 3 is not implicated; an answer blaming it earns no credit.

3. $f(x) = x^2$ with $f^{-1}(x) = \sqrt{x}$ and no restriction. (a) $f(-4)$, then its square root. (b) The restriction.

Solution (a): $f(-4) = (-4)^2 = 16$

$$f(-4) = 16$$

$\sqrt{16} = 4$, not -4 .

$$\sqrt{f(-4)} = 4$$

Solution (b): *Instructor-judged.* The round trip started at -4 and came back at 4 , so $\sqrt{x}$ is not undoing f on negative inputs. The repair is to restrict f to $x \geq 0$, where it is one-to-one; then $\sqrt{x}$ is genuinely its inverse. A student may instead restrict to $x \leq 0$, but only with the inverse changed to $-\sqrt{x}$. Repeating that the numbers disagree is not a diagnosis.

4. Maya wrote $g(x) = 3(|x| - 4)$ for “stretch by 3, then shift down 4”, and reported $g(2) = -6$. (a) Correct rule and $g(2)$. (b) Name her error.

Solution (a): Stretch first, then subtract: $g(x) = 3|x| - 4$, so $g(2) = 3(2) - 4 = 2$.

$$g(2) = 2$$

Solution (b): *Instructor-judged.* Her bracket subtracts 4 and *then* triples, so the 4 was multiplied by 3 and the graph dropped 12 instead of 4. The stretch has to come first, and the -4 must stay outside the multiplication. A correct answer names the order and identifies the 4 as the number that was wrongly multiplied.

5. $f(x) = \sqrt{x - 2}$ with $f^{-1}(x) = x^2 + 2$ on all reals. (a) $f(f^{-1}(-3))$. (b) Why the domain claim fails.

Solution (a): $f^{-1}(-3) = (-3)^2 + 2 = 11$, then $f(11) = \sqrt{11 - 2} = \sqrt{9} = 3$.

$$f(f^{-1}(-3)) = 3$$

Solution (b): *Instructor-judged.* The round trip returned 3, not -3 , so on that input the rule is not an inverse at all. The reason is that f never produces a negative output — its range is $y \geq 0$ — and the domain of f^{-1} must be exactly that range. So $f^{-1}(x) = x^2 + 2$ is correct only on $x \geq 0$. Changing the formula instead of restricting it is not the fix.

6. $f(x) = \sqrt{x}$; a student says $y = f(2x)$ stretches horizontally by 2. (a) Solve $f(2x) = 4$ and $f(x) = 4$. (b) Stretch or compression?

Solution (a): $\sqrt{2x} = 4 \Rightarrow 2x = 16 \Rightarrow x = 8$.

$$x = 8$$

$\sqrt{x} = 4 \Rightarrow x = 16$.

$$x = 16$$

Solution (b): *Instructor-judged.* The height 4 that used to be reached at $x = 16$ is now reached at $x = 8$: every point has moved to half its distance from the y -axis. That is a

horizontal *compression* by a factor of $\frac{1}{2}$, not a stretch by 2. The two computed inputs are the evidence, and an answer that names compression without them is only half the response.

7. $y = -(x+1)^2 + 5$ with the reflection carried out last. **(a)** True vertex height. **(b)** Name the error.

Solution (a): At $x = -1$: $-(-1+1)^2 + 5 = -0 + 5$.

$$y = 5$$

Solution (b): *Instructor-judged.* In the printed rule the minus sign attaches to the squared term alone, so the reflection happens before the $+5$ is added. Reflecting last negates the shift as well and drops the vertex to -5 — the student's number. A correct answer states that a reflection in the x -axis must precede any vertical shift, or equivalently that the minus sign does not reach the $+5$.

8. $f(x) = x^2 - 4x + 7$ on $x \leq 2$; a student used $f^{-1}(x) = 2 + \sqrt{x-3}$ and got $f^{-1}(12) = 5$. **(a)** Correct value. **(b)** Which branch.

Solution (a): Complete the square: $f(x) = (x-2)^2 + 3$. Solving $y = (x-2)^2 + 3$ gives $x = 2 \pm \sqrt{y-3}$, and $x \leq 2$ forces the minus sign, so $f^{-1}(x) = 2 - \sqrt{x-3}$. Then $f^{-1}(12) = 2 - \sqrt{9} = 2 - 3$.

$$f^{-1}(12) = -1$$

Solution (b): *Instructor-judged.* The vertex sits at $x = 2$, and the restriction $x \leq 2$ keeps the *left* branch. Every output of f^{-1} must therefore be at most 2, which is what forces the minus square root; the plus root returns values at or above 2 and inverts the right branch instead. A complete answer names the vertex, says which branch is kept, and ties that to the sign.